

Web Scraping and Job Market Analysis from Wuzzuf

Project Overview

This project involved scraping job posting data from Wuzzuf.net, a popular Egyptian job listing website. The primary goal was to gather real-world data on the Egyptian job market, clean and prepare it for analysis, and extract meaningful insights through data analysis and visualization.

Objectives

- Extract job listings and relevant metadata from Wuzzuf using web scraping.
- Preprocess and clean the data to ensure quality and consistency.
- Apply encoding techniques to prepare categorical variables for analysis.
- Perform exploratory data analysis (EDA) to understand trends in the job market.
- Visualize the data to uncover patterns related to job titles, companies, locations, experience levels, and required skills.

Tools & Technologies Used

- Programming Language: Python
- Libraries: BeautifulSoup, Requests, Pandas, NumPy, Matplotlib, Seaborn, Plotly
- Jupyter Notebook for development and visualization
- CSV/Excel for storing cleaned data

Workflow & Methodology

1. Data Collection:

- Used requests and BeautifulSoup to scrape job data from Wuzzuf.
- Collected fields like job title, company name, location, experience, job type, and skills.

2. Data Cleaning:

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- Removed duplicates and nulls, standardized fields, cleaned text data.

3. Data Encoding:

- Converted categorical data using Label and One-Hot Encoding.

4. Data Analysis:

- Analyzed job distribution, in-demand skills, experience levels, remote roles.

5. Data Visualization:

- Used Seaborn, Matplotlib, and Plotly for visual insights including bar plots, heatmaps, and word clouds.

Key Insights

- Most job opportunities are concentrated in Cairo, followed by Alexandria.
- Most in-demand roles are in software development, sales, and marketing.
- Top required skills include Python, communication, and problem-solving.
- Majority of jobs required 1-3 years of experience.

Conclusion

The project demonstrated how web scraping combined with data analytics can provide valuable insights into the job market. It revealed patterns that can benefit job seekers, recruiters, and career development platforms.

Future Work

- Automate data collection with scheduled scripts.
- Expand to multiple job sites for broader analysis.

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- Use NLP to better extract insights from job descriptions.